

RÉSUMÉ

Alec *Layton.*

Software Engineer · Backend Systems · Automation · AI Agent Tooling

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Software engineer with 3+ years building and owning mission-critical software across the full development lifecycle, with a focus on automation, data pipelines, and microservices. Ships systems that process hundreds of millions of dollars in transactions, including ETL automation that eliminates 100+ hours of manual work per quarter while holding error rates under 1%. Increasingly focused on AI agent tooling — least-privilege automation with human-in-the-loop approval on anything outward-facing. Thrives in fast-moving, high-ownership environments.

EXPERIENCE

Software Engineer

December 2022 – Present

Paycom · Irving, TX

- Architected and implemented an internationalization framework for an enterprise-scale tax API using C# and .NET, enabling automated multi-country payroll controls processing across 5 countries with high compliance accuracy.
- Led end-to-end design and delivery of mission-critical regulatory compliance controls for domestic garnishment, tax withholding, and earnings regulations within the core payroll engine — processing hundreds of millions of dollars in annual transactions while maintaining 100% audit compliance.
- Independently architected and deployed a full-SDLC automated ETL pipeline to parse, validate, and transform 50+ state-specific SUTA tax rate files into controlled system updates, eliminating 100+ hours of manual processing per quarter and reducing error rates to under 1%.
- Proactively monitor production systems, triage incidents, and implement reliability improvements — reducing downtime and increasing application availability for enterprise-scale payroll infrastructure.
- Collaborate cross-functionally with product, QA, and engineering teams to gather requirements, design system architecture, and conduct code reviews that uphold reliability and security standards.

Enterprise Architect Developer — Internship

May 2021 – December 2021

Valero Energy Corporation · San Antonio, TX

- Developed a mission-critical SAP GUI application in ABAP for the Accounts Payable team, supporting processing for 90% of all U.S. financial transactions with high reliability and accuracy.
- Engineered enterprise-grade software within a Fortune 500 environment, translating complex operational requirements into scalable, production-ready systems.
- Documented software components, integration flows, and technical specifications for cross-team knowledge transfer.

PROJECTS

Kowalski Personal AI Agent System

- Architected a multi-tool AI agent (built on Claude Code) integrating 15+ MCP servers spanning Google Workspace, GitHub, Obsidian, and voice synthesis, with persistent structured memory and automated cross-system documentation.
- Designed least-privilege automation workflows including scheduled cron-based routines, security audit protocols, and human-in-the-loop approval gates for external-facing actions such as email, calendar, and code deploys.

AI Gaming Assistant Full-Stack Desktop Application

- Built a cross-platform desktop gaming companion (Tauri 2 + Rust, React/TypeScript; FastAPI backend on Supabase + Upstash Redis) with streaming AI chat, automatic session tracking across 80+ games, Discord integration, and a content feed.
- Engineered a dual-mode AI inference router with failover across cloud providers (Groq, Together AI) and local models (Ollama); shipped Stripe subscription billing and a game-aware notification engine, backed by 238 automated tests across Python, TypeScript, and Rust.

TECHNICAL SKILLS

LANGUAGES	Python, C, C++, C#, JavaScript, SQL, Rust, PHP, ABAP
BACKEND & APIS	.NET, Node.js, FastAPI, RESTful APIs
FRONTEND	React, TypeScript SPAs, Tailwind CSS
DATABASES	PostgreSQL, SQLite, MySQL, query optimization, data pipeline development
DEVOPS & CLOUD	Git, Docker, AWS EC2, Linux/UNIX
PRACTICES	Agile/Scrum, TDD, code reviews, system design, production monitoring

EDUCATION

B.S. Computer Science Graduated December 2022

The University of Texas at San Antonio

Concentrations in Software Engineering & Data Science.

ADDITIONAL

- ACM ICPC member — strengthened algorithmic problem-solving through team-based competitive programming.
- Experienced public speaker with strong written and verbal communication skills.
- Bias toward full ownership and rapid iteration in high-intensity, mission-driven environments.